

Finn Hertsch

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RESEARCH INTERESTS

Representation learning and high-dimensional retrieval, numerically robust online decision-making (contextual bandits), planetary remote sensing data pipelines, and scalable database architectures.

EDUCATION

DHBW Ravensburg (Baden-Wuerttemberg Cooperative State University) 10/2024 – 09/2027 (expected)
B.Sc. in Data Science and Artificial Intelligence (210 ECTS) *Current GPA: 2.3 (CS Core GPA: 1.5)*

- Cooperative dual study program combining rigorous theoretical coursework with applied industry engineering.
- Core Focus: Advanced programming (1.3), theoretical computer science (1.5), machine learning, distributed systems.
- **Project Work I (Projektarbeit 1):** Hybrid article recommendation microservice on Google Cloud Platform with Schwäbisch Media (**Grade: 1.0 / Score: 97/100**).
- *Bachelor Thesis scheduled for final academic year (2027).*

Edith-Stein-Schule Ravensburg

Abitur (General University Entrance Qualification)

2021 – 2024

Final Grade: 2.0

- Focus Subject: Biotechnology.
- Coursework & Practicals: Sequence alignment and database retrieval (BLAST, FASTA), GFP reporter-gene transformation in *E. coli*, fermentative biosynthesis in bench-scale bioreactors.

PUBLICATIONS

- [1] **F. Hertsch**, "*Kairos: Numerically Robust News Recommendation under Item Cold-Start via Cholesky-based LinUCB.*" arXiv preprint [arXiv:2607.26832](https://arxiv.org/abs/2607.26832) [cs.LG], 2026.
German version peer-reviewed and accepted at **SKILL 2026** (16. Studierendenkonferenz Informatik), GI-Edition: Lecture Notes in Informatics (LNI), Gesellschaft für Informatik. Presented at INFORMATIK 2026, TU Dresden, September 2026. LNI proceedings & DOI forthcoming.

RESEARCH EXPERIENCE & ENGINEERING PROJECTS

Kairos: Numerically Robust Contextual Bandits (Julia) 2026
Lead Researcher & Developer / arXiv:2607.26832 / Accepted at SKILL 2026 (LNI)

- **Numerical Stability & Cholesky Updates:** Developed **Kairos** in **Julia**, replacing standard Sherman-Morrison matrix inversion with direct rank-1 updates of Cholesky factors (**cholupdate**) in contextual Linear Upper Confidence Bound (LinUCB) bandits, preserving SPD matrix structure and preventing catastrophic condition number divergence under sparse item cold-start data (TTL < 48h).
- **Monte Carlo Stability Verification:** Validated numerical resilience across 1,000 Monte Carlo simulation runs, proving zero error variance in Cholesky updates compared to explosive divergence in traditional inversion.
- **Adaptive MRL Retrieval Subspace:** Integrated Matryoshka Representation Learning (MRL) for Maximum Inner Product Search (MIPS) candidate retrieval, reducing embedding dimensionality from 768d to 128d to achieve a **4.78× inference speedup** (0.065 ms vs 0.312 ms / 100 items) with an MAE of 0.032 (retaining > 96.8% semantic structure).
- **Streaming Validation:** Deployed and benchmarked using live streaming data from the Tagesschau API.

Production-Grade Hybrid Recommendation Microservice (Project Work I) 2025
DHBW Ravensburg / Schwäbisch Media / Grade: 1.0 (Score: 97/100)

- **Cloud-Native Microservice Architecture:** Engineered an asynchronous FastAPI (ASGI) microservice on Google Cloud Platform orchestrating parallel prediction pipelines across Vertex AI Vector Search (3072d embeddings, MIPS) and Neural Collaborative Filtering (NCF / MLP).
- **Bayesian Hyperparameter Optimization:** Implemented a Tree-structured Parzen Estimator (Optuna TPE) search across 515,000 candidate evaluations (515 trials over 1,000 users), identifying an optimal 0.7226 / 0.2774 CBF-CF score-fusion equilibrium.
- **Rigorously Unbiased Evaluation:** Evaluated performance under a Leave-Last-Out Full-Catalog Ranking protocol over 2.3M users and 104k items (< 0.005% matrix density), proving a **5.4× nDCG@10 improvement** over baseline models at a 95th-percentile latency of 1,562 ms (SLO < 2,000 ms).

Planetary Remote Sensing Representation Learning

Research collaboration with Prof. Dr.-Ing. Mark Schutera, DHBW Ravensburg (Ongoing Research)

2026 – Present

- **Domain Exploration:** Investigating self-supervised representation learning and compact vector embeddings for planetary surface imagery (Lunar Reconnaissance Orbiter / LROC NAC data).
- **Data Processing Pipeline:** Developing ingestion and preprocessing workflows using USGS ISIS3 and PyTorch to support multi-resolution planetary imagery analysis.

PROFESSIONAL EXPERIENCE

Hertsch Technologies UG (haftungsbeschränkt), Germany 02/2026 – Present
Founder & Managing Director

- Founded and lead a software engineering company focused on automated data processing pipelines and maintenance infrastructure.

Waldner Holding, Wangen im Allgäu, Germany 07/2026 – 09/2027
Dual-Study Industry Partner – Data Science & Software Engineering

- Applied data science and systems engineering as part of the DHBW cooperative study program.

Schwäbisch Media (Schwäbischer Verlag), Ravensburg, Germany 10/2024 – 06/2026
Dual-Study Industry Partner – Data Science & Software Engineering

- Developed and deployed a hybrid recommendation microservice (NCF & CBF) on Google Cloud Platform (Vertex AI, BigQuery).
- Designed data ingestion pipelines processing high-volume user interaction logs with BigQuery.

TECHNICAL SKILLS

- **Programming Languages:** Julia, Python, Java, SQL, C/C++.
- **ML & Scientific Computing:** Contextual Bandits (LinUCB), Matryoshka Representation Learning (MRL), Neural Collaborative Filtering (NCF), PyTorch, Optuna (TPE), Vector Search (Vertex AI Vector Search), Julia Scientific Computing (BenchmarkTools.jl).
- **Cloud & Systems Engineering:** Google Cloud Platform (Vertex AI, BigQuery), FastAPI (ASGI), Docker, Linux, Git, USGS ISIS3.

LANGUAGES

German: Native · **English:** Professional / Academic (B2/C1) · **Spanish:** Intermediate (B1)